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From	Our Reference	Pittsburgh, PA
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Robot Learning Research Intern

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The **Robot Learning Lab** at the **Bosch Center for Artificial Intelligence** (BCAI) in **Pittsburgh** invites research interns to help build robots that acquire transferable skills, adapt to unfamiliar situations, and improve through real-world experience. Working at the intersection of robot learning, multimodal machine learning, embodied AI, and generative AI, we turn challenging robotics and automation problems into fundamental scientific investigations with industrial impact. Our goal is to develop reliable robotic systems that augment human capabilities and collaborate with people and other agents.

Research opportunities span:

- **Robot foundation models:** vision–language–action models, test-time adaptation and steering, and learning from deployment experience and failures.
- **Agentic robot systems:** coding agents and multi-agent frameworks for skill discovery, composition, long-horizon planning, and robot control.
- **Dexterous and contact-rich manipulation:** integrating vision, touch, and proprioception for precise interaction with rigid and deformable objects.
- **Transferable robot skills:** cross-embodiment representation learning, learning from human and robot demonstrations, and retrieval-based policy generalization.
- **Simulation and world modeling:** generative digital twins and Real2Sim2Real pipelines for policy learning, evaluation, and transfer to physical robots.

We combine Bosch's industrial research environment with close collaborations across the Robotics Institute at Carnegie Mellon University, including dual affiliations among lab members. Interns help shape research questions, develop new methods, and evaluate systems in simulation and on physical robots, with the aim of publishing at leading robotics and machine learning venues. Across projects, we emphasize rigorous experimentation, effective use of open-source models and tools, and a clear path from scientific insight to practical robotic capability.

Job Description

We seek research interns who combine strong scientific judgment with the initiative to take ownership of an ambitious research project. Working closely with their mentors and collaborators, interns will formulate compelling research questions, critically assess prior work, develop and implement new methods, and design experiments that produce clear, defensible scientific insights.
Successful candidates will bring strong foundations in machine learning and robotics, with depth in one or more of the following areas:

- **Generalization and transfer:** learning multimodal representations and robot policies that transfer across tasks, objects, environments, and embodiments, including settings with limited demonstrations or deployment data.
- **Policy learning and robot foundation models:** imitation learning, reinforcement learning, generative policies, and the training, fine-tuning, or post-training of vision–language–action and other embodied foundation models.
- **Learning at deployment:** test-time adaptation, policy steering, failure detection and recovery, and agentic frameworks for reasoning, skill composition, and continued improvement.
- **Data and simulation:** combining human demonstrations, robot trajectories, and synthetic data; developing simulation or world-model-based pipelines for training, evaluation, and transfer to physical robots.
- **Effective implementation:** translating research ideas into reliable software, implementing and debugging neural policies, and using efficient training, inference, and distributed GPU workflows where appropriate.
- **Rigorous evaluation:** selecting informative baselines and ablations, diagnosing failures and distribution shifts, examining dataset biases, and characterizing robustness, data efficiency, and generalization.

Interns will collaborate with teammates to develop research contributions for publication at leading venues and contribute to invention disclosures and patent applications arising from their work.

Tasks

- Define a focused research problem with mentors and collaborators, connecting an important scientific question to a meaningful robotics or automation challenge.
- Critically review relevant literature, identify limitations in existing approaches, and articulate a testable research hypothesis.
- Develop an R&D plan specifying the proposed methods, data sources, experimental settings, evaluation metrics, milestones, and target publication venues.
- Design and implement the proposed approach, maintaining reproducible code, documented experimental configurations, and clear records of results.
- Evaluate methods in simulation and on physical robots through quantitative comparisons, targeted ablations, qualitative analysis, and theoretical analysis where appropriate.
- Share progress, relevant literature, experimental findings, and open challenges in weekly discussions; refine the research plan as evidence emerges.
- Prepare a research manuscript with collaborators for submission to venues such as CoRL, RSS, ICLR, NeurIPS, ICML, ICRA, IROS, Science Robotics, IJRR, T-RO, CVPR, ICCV, ECCV, etc., and contribute to associated technical reports and patent materials.

Qualifications

Required Qualifications

- Currently pursuing a Ph.D., or a recent postdoctoral researcher, in Robotics, Machine Learning, Computer Science, Computer Engineering, Electrical Engineering, or a closely related field.
- Minimum GPA of 3.0 on a 4.0 scale, or equivalent.

- Strong foundations in machine learning and robot learning, including representation learning, policy learning, optimization, and experimental design.
- Hands-on experience training or adapting modern robot policies and foundation models, such as vision-language-action (VLA) transformers with autoregressive action-token prediction or diffusion/flow-matching action heads for continuous action-chunk generation, or world action models (WAMs) that jointly predict future observations and robot actions. Relevant training experience includes multi-task pretraining or behavior cloning/supervised fine-tuning with action-token cross-entropy, diffusion denoising, conditional flow-matching, or joint video-action prediction objectives; or offline or online reinforcement learning post-training using actor-critic or policy-gradient methods.
- Strong Python programming skills; experience implementing, debugging, and evaluating neural models using PyTorch or a comparable framework; and hands-on experience using coding agents to develop, test, and maintain robotics and machine learning software, including reviewing and validating agent-generated code.
- Practical experience with physical robot platforms, including robot data collection, policy deployment, and closed-loop experimentation.
- Ability to design rigorous evaluations using appropriate baselines, ablations, and diagnostic analyses to characterize generalization, robustness, and failure modes.
- Demonstrated research ownership, supported by publications, preprints, or substantial research projects, together with clear technical communication and effective collaboration.

Desired Qualifications

- Experience training or adapting robot foundation models using heterogeneous human demonstrations, robot trajectories, and synthetic data, including data curation, quality assessment, and cross-embodiment transfer.
- Experience integrating large language models, vision–language models, or other multimodal foundation models into robot perception, planning, control, or evaluation workflows.
- Experience with test-time adaptation, policy steering, embodied memory, failure detection and recovery, or agentic systems that generate, execute, and refine robot skills.
- Experience in dexterous or contact-rich manipulation, including learning from visual, tactile, force, proprioceptive, or acoustic feedback.
- Experience with simulation, generative world models, digital twins, or Real2Sim2Real pipelines for policy training, evaluation, and transfer.
- Experience with multi-GPU or multi-node training, efficient fine-tuning and inference, and reproducible workflows encompassing versioned datasets, experiment tracking, and checkpoint management.
- Proficiency developing robotics and machine learning software on Linux, using version control and maintainable software practices.
- Theoretical depth in areas such as transfer learning, reinforcement learning, optimization, uncertainty estimation, or safe learning and control.
- A strong publication record at leading robotics, machine learning, or computer vision venues, particularly contributions demonstrating methodological depth and careful empirical validation.